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On the predictive validity of the National Football League combine: does it forecast future success?

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ABSTRACT

The National Football League (NFL) Combine provides NFL teams the ability to assess prospective athletes' medical histories and physical and psychological abilities. Using this information, NFL personnel must then decide whether an athlete is a good fit to their team. Given the combine's 40-year history and the availability of peer reviewed articles on the function and efficacy of the combine, the purpose of this systematic review was to synthesize the literature evaluating the predictive validity of the combine according to PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses). Full-text, peer-reviewed articles containing information relevant to the NFL combine and at least one measure of future success were retained. The search yielded 1954 articles and after screening, 68 articles remained. These remaining articles focused on measures of success pertaining to (a) medical testing and future performance ($n = 25$), (b) combine tests ($n = 12$), (c) draft position ($n = 10$), (d) draft and future performance ($n = 8$), (e) medical testing ($n = 7$), (f) career longevity ($n = 3$), (g) draft and salary ($n = 2$), and (h) playing performance ($n = 1$). Due to the mixed results of combine measures on future success, this review highlights the need for more research to investigate the combine's influence on long-term performance and success.

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KEYWORDS
Scouting; performance; talent identification; athlete selection; injury; amateur draft

1. Introduction

In 1982, National Football Scouting Incorporated held the first-ever National Football League (NFL) combine in Tampa, Florida (National Football Scouting. NFL Combine History, 2020). The combine is held annually to assess draft-eligible, college football prospects on a variety of physical, physiological, and psychological variables. Additionally, athletes undergo a series of medical examinations to estimate the risk of future injury and to inform player training/development programming (National Football Scouting. NFL Combine History, 2020). An athlete's performance at the combine is believed to hold important insight into his[1] position in the draft which also has important implications on the athlete's starting salary (Treme & Allen, 2011). Like the combine, the NFL draft is a formal, annual event designed to increase the competitive parity between the teams in the league whereby players are drafted to join a particular NFL team. Presently, the draft consists of seven rounds whereby each of the 32 teams takes turns selecting prospects in a predetermined order based on the team's performance in the previous year. In this sense, the team with the lowest ranking is assigned the first draft pick while the team that won the Super Bowl championship is assigned the final pick position (NFL Football Operations). Ahead of the draft, administrative staff have collected a significant amount of data on draft-eligible prospects using a variety of information sources (interviews, video footage, performance metrics) including the information gleaned from the player combine. While not all teams value and weigh the player combine equally when judging a player's likelihood of success, the combine is still regarded as an important component in the decision-making strategy for many (Berri & Simmons, 2011).

For this reason, the combine is designed in a way that allows for the evaluation of each draft-eligible prospect in a standardized setting (Kuzmits & Adams, 2008). In this regard, athletes perform a series of tests within the same environment that subjects them to uniform evaluation measures (e.g., resources such as equipment, facilities, raters/evaluators are as consistent and reliable as possible). While the psychological, medical, and interview evaluations are not broadcasted on television for public viewing, the physical battery of tests is, which is what many come to associate the combine with. Composed of six tests, the physical battery includes the 40-yard dash with 10 and 20-yard split times recorded, vertical jump, horizontal jump, 20-yard shuttle, 3-cone drill, and 225-pound bench press (Robbins & Goodale, 2012). While the 10 and 20-yard split times are measures of acceleration, the 40-yard dash provides a measure of a player's maximum speed. Both the vertical and horizontal jumps are administered as measures of lower-body strength and power. The 20-yard shuttle and 3-cone drill are nonreactive measures of change-of-direction ability. Lastly, the 225-pound maximum repetition bench press test is purported to be a measure of upper-body strength. However, since players perform repetitions ranging anywhere from 16 to 29 in the centre and defensive tackle positions, respectively, the 225-pound bench press test is often regarded as a measure of upper-body endurance. It is important to note that not all positions are subject to the same battery of tests. For example,

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[1]Henceforth, him/he/his will be used to refer to the male participants at the NFL combine.

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quarterbacks and wide receivers are exempt from the bench press test, while kickers and punters solely perform the 40-yard dash (Robbins & Goodale, 2012).

Along with physical testing, players complete the Wonderlic test, a test designed to measure general intelligence (Mirabile, 2005). This cognitive test was developed by Eldon F. Wonderlic in 1936 and consists of 50 multiple-choice questions performed within the span of 12 minutes (Mirabile, 2005). The predictive validity of the Wonderlic has been called into question with mixed results being reported across numerous studies. For instance, Gill and Brajer (Gill & Brajer, 2012) found the Wonderlic test to be important in explaining draft selection for quarterbacks, offensive linemen, and tight ends. Contrarily, Pitts and Evans (Pitts & Evans, 2018) found the Wonderlic scores to not be significant predictors of "productivity"[2] among NFL quarterbacks. The use of heterogeneous dependent variables between these studies (i.e., draft selection by Gill and Brajer (Gill & Brajer, 2012) and NFL performance within the first 3 years by Pitts and Evans (Pitts & Evans, 2018)) to measure the predictive validity of the Wonderlic may account for these differences. Meanwhile, other studies have found the Wonderlic test to have no correlation or bearing on NFL success (Pitts & Evans, 2018; Mirabile, 2005).

In addition to concerns with the predictive validity of the Wonderlic test, the physical battery has also been found to have mixed results. McGee and Burkett (McGee & Burkett, 2003) found measures of height, weight, the 3-cone drill, the vertical jump test, and 10-yard sprint (split time) to be significant predictors of draft order for running backs, wide receivers, and defensive backs. Meanwhile, Tucker and Black (Tucker & Black, 2021) found the 3-cone drill had no correlation with improving draft position and concluded that combine performance is not a good predictor of draft position except possibly for wide receivers.

Perhaps in relation to the questionable efficacy of the combine in predicting future NFL success, it appears there is much room for improvement (Baker & Wattie, 2018; Johnston et al., 2022; Koz et al., 2012). One of the most famous examples to call into question the predictive validity and inexact science of the combine is New England Patriots' quarterback, Tom Brady: nearly the slowest of his draft class in the 40-yard dash, and 199th pick of his year's draft, but has since gone on to win multiple Super Bowls and Most Valuable Player awards (Johnston et al., 2022; Kacsmar, 2013). Given that scientific inquiry into combine tests is limited, it raises concerns about the combine's validity (Kuzmits & Adams, 2008). This is especially true because NFL teams do not publicly disclose how they use combine measures to inform draft selection due to proprietary reasons and other data-sharing concerns which leaves sport science researchers to rely on performance metrics, salary, and draft status, among other variables, to infer the predictive validity of the NFL combine. Moreover, of the research that does exist, mixed findings leave researchers and practitioners questioning the validity of some tests over others. For example, Pollock et al (Pollock et al., 2021) found the 20-yard shuttle run to be significantly associated with 1-year and 5-year roster status for linebackers and inside defensive linemen. In contrast, Vincent et al (Vincent et al., 2019) found the 20-yard shuttle run had no significant correlation with any performance variable except for linebacker's solo tackles.

For these reasons, the purpose of this review is to capture the current body of evidence evaluating the efficacy of the combine (as described by the study) using various dependent variables. While previous systematic reviews and meta-analyses have explored the efficacy of the combine with respect to singular dependent variables such as salary earnings and games started, this review adds novelty to the literature by providing a comprehensive overview that synthesizes all the dependent variables available through our search strategy, which was used to evaluate measures of combine predictive validity for future NFL success. The broad objectives of this review are to (a) better understand what dependent variables have been explored in this field of research, (b) summarize the constituent findings in a relevant way, (c) identify gaps in the literature, and (d) inform future research to address these gaps as well as other over/underestimates of knowledge. The benefit of such research may positively (a) inform the degree of reliability and validity of the combine as a predictive measure of talent identification (b) inform evidence-based models of combine development, (c) identify gaps in current understanding and directions for future work, and (d) highlight potential areas of future research foci to address the identified gaps (i.e., career longevity, draft position, playing performance, etc.) and position-specific information across studies regarding the NFL combine.

2. Methods

2.1. Key search terms and search strategy

The present review was conducted according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines (Moher et al., 2009; Tricco et al., 2018). The search strategy consisted of two phases, per the PRISMA guidelines. The first phase included searching through five databases: EconLit, PubMed, Sports Medicine and Education Index, Sport Discus and Web of Science due to their robust catchment of sports science journals. The following search terms were used to identify the abstract and title: (a) "combine" AND "sport", and (b) "combine" AND "NFL". To determine the specific search terms, the authors completed a preliminary scoping review of the research to identify common language used in this field of research which revealed there are few terms for the "combine". Due to its specific nature, the word "combine" was chosen as there are no other true names for its description in North America. Phase two consisted of a secondary search through external sources (e.g., reference lists of articles found in phase one) to yield a greater number of articles. Upon completion of these two phases, the study's author(s), title, and year of publication were recorded and duplicates were removed across the databases.

2.2. Inclusion exclusion criteria

From the remaining articles, the title and abstract were read to discern whether they were eligible for further screening if they (a) were in full-text journal article format (i.e., theses, dissertations, conference proceedings, and commentaries were not

[2]Productivity is measured as the number of games started per season, approximate value per season wins per season and passing yards per season.

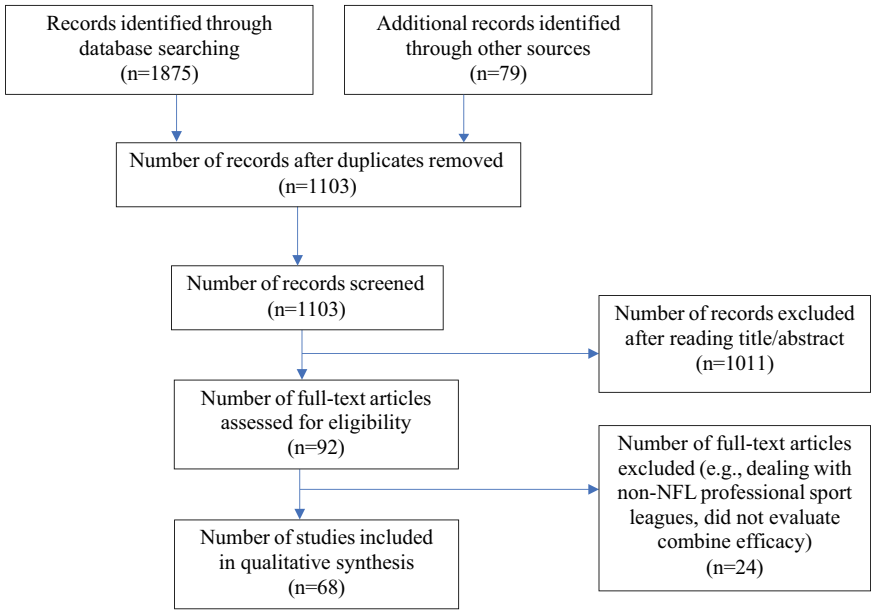

Figure 1. PRISMA flow chart depicting the number of records collected, screened and the final number of articles included for analysis.

included), (b) published between January 1900 and September 2021, (c) written in English, (d) related to sport, and (e) specifically examined the combine, athlete selection or, identification processes. Following the initial screening, a secondary screening was conducted to determine whether the articles met the final inclusion criteria of (a) focused on the North American, men's professional National Football League and (b) examined the efficacy of the combine (e.g., how the combine relates to some measure of success in the future). A full-text analysis of the remaining 68 articles was then conducted. The PRISMA process and the associated number of articles are depicted in Figure 1.

2.3. 2.3 Groupings

Each article retained after the final screening was coded based on the following criteria: publication year, sample size (<20 participants, 20–50 participants, 51–100 participants, 101–200 participants, 201–500 participants, >500 participants) and positional information (defence only, offence only, both, position-specific or none). In addition, articles were thematically coded according to the measure(s) of combine success including (a) career longevity, (b) draft position, (c) combine tests and future performance, (d) medical testing, (e) playing performance, (f) combination of draft and future performance, (g) combination of draft and salary, and (h) combination of medical testing and future performance. These foci were strategically selected based on criteria and future outcomes used by Johnston et al (Johnston et al., 2022).

2.4. Reliability of coding

To ensure the reliability of coding for each article, a sub-sample of 103 randomly selected articles were evaluated independently by the secondary author at the start of the coding process and produced 100% inter-rater reliability based on the determined inclusion and exclusion criteria.

3. Results

Based on the initial data collection phase, 1875 articles were extracted through database searches with an additional 79 extracted from outside these databases (i.e., using reference lists and additional readings). After the removal of duplicates, 1103 articles remained and were screened based on their title and abstract which led to the removal of 1011 articles. Of the remaining 92 full-text articles, 68 met the final inclusion criteria and were considered for analysis as the others examined professional sports leagues other than the NFL and/or did not examine at least one measure of the combine's predictive validity (see Figure 1). Table 1 presents descriptive statistics summarizing the number of studies published according to year, sample size, research foci, and positional information.

3.1. Publication year

The first study evaluating the predictive validity of the NFL combine can be found dating back to 2003. Since then, there has been growing interest in this area of research as seen in Table 1. While there is a clear spike in the number of published studies examining the predictive validity of the NFL combine in 2018, 81% of studies were published within the last decade, indicating growing interest in this research area.

3.2. Sample size

Sample sizes ranged from 14 participants (Vopat et al., 2017) to 5506 participants (Hedlund, 2018). Most studies (50%) employed larger sample sizes exceeding 500 participants. This is a promising outcome as large sample sizes enhance the reliability of findings and more closely reflect the hundreds of diverse participants invited to participate in the NFL combine annually.

Table 1. Descriptive statistics according to publication year, sample size, foci, and positional information.

Year	N	%
2003	1	1.5%
2004	1	1.5%
2005	1	1.5%
2008	3	4.4%
2009	3	4.4%
2010	4	5.9%
2011	8	11.8%
2012	4	5.9%
2013	3	4.4%
2014	3	4.4%
2015	2	2.9%
2016	2	2.9%
2017	7	10.3%
2018	17	25.0%
2019	3	4.4%
2020	4	5.9%
2021	2	2.9%
Category	N	%
Sample Size		
<20	2	2.9%
20–50	5	7.4%
51–100	8	11.8%
101–200	9	13.2%
201–500	10	14.7%
>500	33	48.5%
Not Reported	1	1.5%
Focus		
Combine Measures/Testing		
Career Longevity	3	4.4%
Combination of draft and future	8	11.8%
Combination of draft and salary	2	2.9%
Combine Tests	12	17.6%
Draft Position	10	14.7%
Playing Performance	1	1.5%
Medical History/Testing		
Combination of medical testing and future performance	25	36.8%
Medical Testing	7	10.3%
Positional Information		
Defense	1	1.5%
Offense	11	16.2%
Both	7	10.3%
Categorization	45	66.2%
None	4	5.9%

3.3. Positional information

Positional information was evaluated to better understand the position-specific physical and mental demands placed on each player (Robbins & Goodale, 2012). Of the 64 studies that examined the position-specific demands of the NFL combine, one looked at the demands placed on defensive positions only, 11 looked at the demands placed on offensive positions only, and seven examined the demands placed on both offensive and defensive positions (see Table 1). Yet, most studies (66%) evaluated the predictive validity of the combine according to categorization based on specific player positions.

3.4. Research foci

Through the process of thematic analysis, the authors created two general groupings based on the studies' main research foci: medical history/testing and combine measures/testing. Combine measures and testing encompassed career longevity, draft and future performance, draft and salary, combine tests and future performance, draft position, and playing performance. The vast majority of studies examined medical testing at the combine and future performance ($n = 25$ studies, 36.8%), followed by combine tests ($n = 12$, 17.6%), draft position ($n = 10$, 14.7%), and a combination of draft and future performance ($n = 8$, 11.8%) (see Table 1). Due to the wealth of data, it was warranted to break down the findings of studies examining medical testing and future performance, combine tests, draft and future performance, as well as draft position into Tables 2, 3 , 4 and 5, respectively.

3.5. Predictive validity of medical history

The 25 articles examining medical testing at the combine with future performance looked specifically at Lisfranc injuries ($n = 2$), scaphoid repair ($n = 1$), cervical/lumbar spine injuries ($n = 3$), rotator cuff tears and repairs ($n = 3$), superior labrum anterior and posterior (SLAP) repair ($n = 2$), anterior cruciate ligament (ACL) injuries ($n = 5$), posterior cruciate ligament (PCL) injuries ($n = 1$), meniscectomy ($n = 3$), anterior and posterior labral tears ($n = 3$), Jones fractures ($n = 3$), hip arthroscopic procedures ($n = 1$), knee chondral injuries ($n = 1$), glenoid augmentation ($n = 1$), athletic pubalgia ($n = 1$) and navicular injury ($n = 1$). While a thorough examination of each injury's impact on future

performance is beyond the scope of this paper, some consensus between injuries was found.

3.5.1. Lisfranc injuries

Both McHale et al (McHale et al., 2017) and Wang et al (Wang et al., 2018) found that Lisfranc injuries resulted in an inferior draft position, shortened career length and fewer games played and started during prospects' early NFL careers.

3.5.2. Rotator cuff tears and repairs

Rotator cuff tears and repairs had similar adverse effects such as shortened careers, reduced number of games started and played in a prospect's early NFL career, and a significantly reduced likelihood of being drafted (Gibbs et al., 2016; Wang et al., 2018).

3.5.3. ACL and PCL injuries

With orthopaedic injuries to the knee being extremely common in the NFL, Wang et al (Wang et al., 2018), Provencher et al (Provencheret al., 2018), Keller et al (Keller et al., 2015) and Brophy et al (Brophy et al., 2009; Brophy et al., 2008) concluded that ACL injury and reconstruction resulted in a significantly worse mean draft pick number and draft round pick, starting and playing significantly fewer games, a significantly lower snap percentage, and a shortened career length for offensive and defensive linemen. Meanwhile, isolated PCL injuries had no effect on draft position, or the number of games played (Chahla et al., 2018). In accordance with Chahla et al (Chahla et al., 2018), Logan et al (Logan et al., 2018) noted that while Grade I and II PCL injuries did not affect overall mean draft pick, players with Grade III PCL injuries were more likely to go undrafted. Positions most susceptible to PCL injuries included running backs and offensive linemen, followed by defensive linemen.

3.5.4. Jones fractures

The effect of Jones fractures on future performance had mixed results. While Tu et al (Tu et al., 2018) found players suffering from Jones fractures not to be at risk of going undrafted or playing and starting fewer games during their first NFL season, Spang et al (Spang et al., 2018) reported these athletes had a worse draft pick number, started fewer games, and had overall poorer NFL performance (see Table 2).

3.5.5. Medical testing

The seven studies evaluating medical testing at the NFL combine examined PCL injuries, muscular imbalances, concussion history, cervical spondylosis, and Achilles tendon injury. A player's concussion history had no impact nor relation to his performance on the Wonderlic test (Solomon & Kuhn, 2014). When assessing NFL prospects' height and weight, Hirshorn et al (Hirshorn et al., 2010) found a larger BMI to statistically increase the risk of chondral injuries in quarterbacks, offensive tackles, and defensive ends while linebackers were at greatest risk for cartilage lesions. Linebackers and running backs, two positions prone to the heaviest contact had the lowest MSCSAC values which increased their risk of stingers and cervical spondylosis (Presciutti et al., 2008). Yet, how these injuries impacted players' draft status or future playing performance was not reported. Lastly, defensive NCAA Division I football players who previously sustained an Achilles Tendon injury were less likely to be drafted into the NFL (Wise et al., 2020).

3.6. Predictive validity of combine measures and testing

3.6.1. Combine tests and future performance

The 12 papers evaluating NFL combine tests had mixed results (see Table 3). While LaPlaca and McCullick (LaPlaca & McCullick, 2020) found quarterbacks with a longer broad jump, faster 40-yard dash and 20-yard shuttle run times to play more games during the NFL season, Vincent et al (Vincent et al., 2019) found the vertical jump to also be predictive of NFL playing performance for quarterbacks. Further mixed results were found for tight ends as LaPlaca and McCullick (LaPlaca & McCullick, 2020) reported taller tight ends played more games, lighter tight ends had better NFL performance, the vertical jump correlated weakly with a better grade for pass route[3] while the 40- yard dash, broad jump and vertical jump correlated significantly to having a better longest catch. Meanwhile, Robbins (Robbins, 2011) found only one combine test, the 40-yard dash, to discern draft status for tight ends.

3.6.2. Combination of draft and future performance

The eight papers evaluating the impact of the combine's physical testing on draft selection and future NFL performance had mixed results (see Table 4). For example, while Wolfson et al (Wolfson et al., 2011) reported the 40-yard dash to have no association with the number of games played during the NFL season for quarterbacks, Pitts and Evans (Pitts & Evans, 2018) found the 40-yard dash to have important implications for games played per season, yards per season and approximate value per season for quarterbacks. In the findings for the work by Pitts and Evans (Pitts & Evans, 2018), the authors found no statistically significant evidence that a quarterback's Wonderlic score, or his height improves his draft position. Contrarily, Berri and Simmons (Berri & Simmons, 2011) found that a quarterback's Wonderlic score, height and 40-yard dash time influence draft decision-making since taller, smarter, and faster quarterbacks had a better overall draft pick.

3.6.3. Draft position

Of the 10 studies examining the predictive validity of the combine tests in determining an NFL prospect's draft position, all 10 noted the significance of a faster 40-yard sprint time being associated with improved draft selection (see Table 5). However, there were still mixed results in relationship of other combine tests. For example, Tucker and Black (Tucker & Black, 2021) found the 40-yard dash and bench press test to be statistically significant in determining draft order for all positions except for offensive guards, offensive linemen, offensive tackles, defensive tackles, inside linebackers, outside linebackers, and punters. Meanwhile, Sierer et al (Sierer et al., 2008) found the 40-yard dash to be predictive of draft status across all positions while the bench press test was only significant in

[3]Grade for pass route is a rating determined by multiple Pro Football Focus experts based on the wide receiver/tight end's receiving performance.

Table 2. Characteristics of the studies evaluating medical testing and future performance.

Authors	Injuries	Variable under examination	Key findings
McHale et al (McHale et al., 2017)	Lisfranc	Draft position	Worse
		Career length	Worse
		Games played/started	Worse
Wang et al (Wang et al., 2018)		Games played/started	Significantly fewer games started in early NFL career
Knapik et al (Knapik et al., 2019)	Scaphoid repair	Draft status	No effect during first NFL season
		Games played/started	No effect during first NFL season
Wang et al (Wang et al., 2018)	Cervical/lumbar spine injury	Games played/started	Significantly fewer games started in early NFL career
		Career length	Decreased
Schroeder et al (Shroeder et al., 2015)	Lumbar spine diagnoses	Draft status	Significantly less likely to be drafted
		Career length	Decreased
		Career performance	No significant difference
Paxton et al (Paxton et al., 2010)	Hyperconcavity of lumbar spine endplates	Draft status	No significant difference
		Career length	No significant difference
Wang et al (Wang et al., 2018)	Rotator cuff repair	Games played/started	Significantly fewer games started in early NFL career
		Career length	Decreased
Gibbs et al (Gibbs et al., 2016)	Rotator cuff tears	Draft status	Significantly less likely to be drafted
		Games played/started	Significantly fewer games started and played
		Career length	Played fewer years
Brophy et al (Brophy et al., 2008)	Rotator cuff injury	% defensive linemen playing in the NFL	Less
Wang et al (Wang et al., 2018)	SLAP repair	Games played/started	Significantly fewer games started in early NFL career
Chambers et al (Chambers et al., 2017)		Draft status	No significant difference
		Mean performance score	
		Career length	
		Games played/started	
Wang et al (Wang et al., 2018)	ACL reconstruction	Games played/started	Significantly fewer games started in early NFL career
Brophy et al (Brophy et al., 2009)		% of linebackers playing in the NFL	Lower
		Career Length	Lower career longevity for offensive and defensive linemen
	ACL injury	Draft status	Less likely to play in the NFL
Provencher et al (Provencher et al., 2018)	ACL reconstruction	Draft pick number	Significantly worse mean draft pick number
		Draft round	Significantly worse mean draft round
		Games played/started	Significantly fewer
		Snap %	Significantly fewer
Keller et al (Keller et al., 2015)		Athletic performance at the NFL combine	NFL-calibre athletes who fully recovered and played at the highest level did not have any diminished functional athletic performance
Chahla et al (Chahla et al., 2018)	PCL	Games played/started	Isolated PCL injuries didn't affect the number of games played
		Draft position	Isolated PCL injuries didn't affect draft position but surgically managed athletes started significantly fewer games
Wang et al (Wang et al., 2018)	Meniscectomy	Games played/started	Fewer games played in the first 2 NFL seasons after the combine for offensive linemen, running backs, wide receivers, defensive backs, defensive linemen and linebackers
		Snap %	Lower snap % for offensive linemen, defensive backs, defensive linemen and linebackers
		Draft status	Decreased chance of playing in the NFL for defensive backs
		Career Length	Shorter playing career for defensive backs
Chahla et al (Chahla et al., 2018)		Career performance	Significantly worse performance in the first 2 seasons
		Games played/started	Fewer total games played and started
		Snap %	Significantly lower snap % during the first 2 seasons
Brophy et al (Brophy et al., 2009)	Meniscal injury	Draft status	Significantly affected the ability to play in the NFL and reduced the % of athletes playing in the NFL in all positions except running backs
		Career length	Significantly affected career longevity for defensive secondaries
Murphy et al (Murphy et al., 2018)	Anterior labral tears	Draft position	Did not perform significantly worse than control group
		Games played/started	
		Snap %	
Knapik et al (Knapik et al., 2018)	Labral repair	Draft status	No significant difference in the season after the combine
		Games played/started	

(*Continued*)

Table 2. (Continued).

Authors	Injuries	Variable under examination	Key findings
Murphy et al (Murphy et al., 2018)	Posterior labral tears	Draft position	Significantly affected NFL draft position (selected later) with linemen being drafted significantly later than healthy controls
		Games played/started	Significantly fewer games played during the first 2 seasons
		Snap %	Lower snap % in the first NFL season
Spang et al (Spang et al., 2018)	Jones fractures	Games played/started	Started fewer games
		Draft pick number	Drafted later
		Overall performance	Overall poorer NFL performance
Tu et al (Tu et al., 2018)		Draft status	Not at decreased risk of going undrafted
		Games played/started	Not at decreased risk of playing or starting fewer NFL games compared to athletes without a history of Jones fracture repair during the first NFL season
Carreira, D. S. and Sandilands, S. M. (Carreira & Sandilands, 2013)	Jones/Diaphyseal Stress Fractures	Draft status	Diaphyseal stress fractures didn't affect draft status
		Total number of games played	Diaphyseal stress fractures didn't affect the total # of games played in the NFL
		Career length	Decreased NFL career length in players with previous proximal 5th metatarsal fractures but not statistically significant
Knapik et al (Knapik et al., 2018)	Hip arthroscopic procedures	Draft status	No significant difference
		Games played/started	
Provencher et al (Provencher et al., 2018)	Knee chondral injury	Draft pick number	Previous untreated knee chondral injuries associated with poorer overall draft pick
		Games played/started	Previous untreated knee chondral injuries associated with playing and starting fewer games during the first 2 NFL seasons
Knapik et al (Knapik et al., 2017)	Glenoid augmentation	Draft status	Drafted into the NFL at a lower rate
		Games played/started	No effect during first NFL season
Knapik et al (Knapik et al., 2017)	Athletic pubalgia surgery	Draft status	No significant difference
		Games played/started	
		NFL seasons played	
Vopat et al (Vopat et al., 2017)	Navicular injury	Draft status	Athletes with a navicular injury were 2x as likely to go undrafted
		Career length	Athletes with a navicular injury were less than half as likely to complete 2 NFL seasons

Note: *SLAP= Superior labrum anterior to posterior.
*ACL=Anterior cruciate ligament.
*PCL=Posterior cruciate ligament.
*NFL=National Football League.

determining draft status among centres, offensive guards, offensive tackles, and defensive tackles. Mixed results were also found across the broad jump as McGee and Burkett (McGee & Burkett, 2003) reported the broad jump to be the best predictor of draft success for offensive and defensive linemen while Sierer et al (Sierer et al., 2008) found the broad jump to have no significance in determining draft status for any position.

3.6.4. Career longevity

The three studies examining career longevity each assessed a different independent variable, with Pollock et al (Pollock et al., 2021) examining the influence of combine tests on career longevity while Brophy et al (Brophy et al., 2009, 2011) investigated the impact that certain medical injuries had on players' career length.

Among defensive positions, the 20-yard dash had the most significant associations with 1-year roster status while the shuttle run had significant associations with 5-year roster status[4] for defensive linemen, cornerbacks, linebackers, and safeties (Pollock et al., 2021). More specifically, the 10, 20 and 40-yard dash, 3-cone drill and 20-yard shuttle times correlated with 1-year and 5-year roster status for linebackers while a heavier weight and longer broad jump were only associated with 1-year roster status. For cornerbacks, Pollock et al (Pollock et al., 2021) found the 20- and 40-yard dash times to be associated with both 1- and 5-year roster status, the broad jump to be the only combine measure significantly associated with 1-year roster status while weight, 3-cone drill and 20-yard shuttle drill times were significantly associated with 5-year roster status. For defensive linemen, weight and the 10-yard dash split time were associated with 1-year roster status, the 40- yard dash time was associated with 5-year roster status and the 20-yard shuttle drill was the only combine measure found to be significantly associated with both 1 and 5-year roster status (Pollock et al., 2021).

The final position evaluated by Pollock et al (Pollock et al., 2021) was safeties, whereby an inverse relationship between height and 5-year roster status was found such that shorter safeties had "better" roster status (i.e., each 0.25-inch increase

[4]Roster status refers to the influence of each performance metric on whether a player was on a roster after the combine.

Table 3. Characteristics of the studies evaluating the combine tests.

Authors	Position	Variables	Key findings
LaPlaca and McCullick (LaPlaca & McCullick, 2020)	Quarterback	40-yd	QBs with faster 40-yd dash, longer broad jump and faster shuttle runs played more games
		Broad jump	
		20-yd shuffle	
	Fullback	20-yd shuttle	Fullbacks with faster shuttle runs and lighter in weight had better career performance
		Weight	
	Running back	3-cone drill	Correlate with improved NFL performance
		40-yd dash	
		Height	Shorter and lighter running backs were superior in NFL rushing performance
		Weight	
	Wide Receiver	Vertical jump	Significantly correlated to NFL player performance
		Shuttle run	Significantly correlated to grades for offence and grades for pass route
		Height	Shorter WRs had more correlations with NFL player performance indices
	Tight ends	Broad jump	Correlated to having a better longest catch
		40-yd dash	
		Vertical jump	
		Height	Taller tight ends played more games
		Weight	Lighter tight ends performed better
	Offensive tackle/ linemen	Bench press	Significant correlations with grades for pass block and associated with a better grades for offence and less hurries allowed
		40-yd dash	Significant correlations with grades for pass block
		Height	Taller offensive tackles had more sacks allowed and hurries allowed
	Guards	Shuttle run	Correlated with a better grades for offence
		Vertical jump	Guards with a higher vertical jump and longer broad jump played more games
		Broad jump	
	Centers	Shuttle run	Centers with a better shuttle run had a better grades for pass block, more hurries allowed and more games played
		3-cone drill	Centers with a faster 3-cone drill and longer broad jump had more hurries allowed
		Broad jump	
		Weight	Those heavier in weight had more pressures allowed
	Cornerback	Weight	Heavier cornerbacks allowed fewer yards after catch per pass coverage snap count and was related to a cornerback's ability to tackle
		Height	Taller cornerbacks allowed fewer yards after catch per pass coverage snap count
		Vertical jump	A better vertical jump was correlated with a cornerback's ability to tackle
		40-yd dash	Strong correlation between a faster 40-yd dash and more games played, more receptions allowed and penalties committed per snap count
	Free safety	40-yd dash	Correlated with better pass coverage defence
		Vertical jump	Superior vertical and broad jump strongly correlated with more tackles per snap count, weakly associated with more stops per total snap count
		Broad jump	
		Bench press	Associated with having more stops per total snap count
		3-cone drill	
	Strong safety	Height	Favored taller and heavier strong safeties for their ability to tackle
		Weight	
		Bench press	Better bench press associated with better grades for run defence
		3-cone drill	Better 3-cone drill and 20-yd shuttle run related to better pass coverage
		20-yd shuttle	
	Outside linebacker	40-yd shuttle	Significantly correlated to NFL player performance
		20-yd shuttle	
		3-cone drill	Faster 3-cone drill associated with better grades for overall defence and pass coverage defence
	Inside linebacker	Bench press	Associated with the greatest number of relevant significant correlations to NFL player performance
		Vertical jump	Weakly but significantly correlated to overall grades for defence
		Broad jump	
		40-yd dash	Faster 40-yd dash time correlated with more tackles and sacks
	Defensive end	Vertical jump	Correlated with NFL player performance
		Shuttle run	
		3-cone drill	Strong correlation to a defensive end's ability to rush the passer
		Broad jump	
		40-yd dash	
	Defensive tackle	Broad jump	Correlated with more pressures per pass rush snap count, sacks per pass rush snap count, and hits per pass rush snap count.
		3-cone drill	
		Shuttle run	
		40-yd dash	Strong significant correlations with the grades for pass rush defence, pressures, sacks, hits, and hurries
Vincent et al (Vincent et al., 2019)	Linebacker	20-yd shuttle	Not significantly correlated with any performance variable except for linebacker sacks
	Quarterback & defensive end	Vertical jump	Predictive of NFL playing performance
		Broad jump	
	Defensive tackle & defensive end	Broad jump	Related to sacks

(*Continued*)

Table 3. (Continued).

Authors	Position	Variables	Key findings
Hedlund (Hedlund, 2018)	Quarterback	Height Weight 40-yd dash Vertical jump Bench press Shuttle run 3-cone drill	The most elite QBs (won Pro Bowl or All-Pro awards) are taller, heavier, run the 40-yd dash faster, have a higher vertical jump, bench press more reps, rand run the shuttle run and 3-cone drill slower
	Running back	Weight 40-yd dash Vertical jump Shuttle run 3-cone drill	Elite running backs (won Pro Bowl or All-Pro awards) tend to weigh less, run the 40-yd dash and 3-cone drill faster and have a shorter vertical jump
	Wide receiver	Height Weight 40-yd dash Vertical jump 3-cone drill	Elite wide receivers (won Pro Bowl or All-Pro awards) tend to be taller, heavier, run the 40-yd dash and 3-cone drill faster and have a higher vertical jump
	Tight ends	Height Weight 40-yd dash Vertical jump Bench press Shuttle run 3-cone drill	Elite tight ends (won Pro Bowl or All-Pro awards) tend to be taller, heavier, run a faster 40-yd dash and 3-cone drill, have a higher vertical jump, run a slower shuttle run and complete more bench press reps
	Offensive linemen	Vertical jump Bench press Shuttle run 3-cone drill	Elite offensive linemen (won Pro Bowl or All-Pro awards) have a higher vertical jump, complete more bench press reps, and run a faster shuttle run and 3-cone drill
	Defensive linemen	40-yd dash Vertical jump Bench press 3-cone drill	Elite defensive linemen (won Pro Bowl or All-Pro awards) run the 40-yd dash and 3-cone drill faster, have a higher vertical jump and bench press more reps
	Linebacker	Weight 40-yd dash Vertical jump Bench press Shuttle run	Elite linebackers (won Pro Bowl or All-Pro awards) weigh more, run the 40-yd dash and shuttle run faster, have a higher vertical jump and complete more bench press reps
	Defensive back	Weight Vertical jump Bench press Shuttle run	Elite defensive backs (won Pro Bowl or All-Pro awards) tend to be heavier, have a higher vertical jump, complete fewer bench press reps and run the shuttle run slower
Robbins (Robbins, 2011)	Quarterback Running back Tight ends Wide receiver	40-yd dash	Performance in the 40-yd dash was superior for drafted QBs, RBs, tight ends, WRs and cornerbacks
	Cornerback	Bench press	Drafted cornerbacks had the lowest bench press reps compared to all positions
		Broad jump	Drafted cornerbacks performed better than all positions in the broad jump except for strong safeties and wide receivers
	Wide receiver	3-cone drill	Drafted wide receivers had the fastest time in the 3-cone drill compared to all other player positions
	Defensive tackle	Bench press	Drafted defensive tackles performed the greatest number of bench press reps compared to all other positions
	Free safety	Vertical jump	Free safeties performed better than all other positions in the vertical jump, except for cornerbacks and strong safeties
		Broad jump	Drafted free safeties had a superior broad jump
	Offensive guard	40-yd dash Vertical jump Broad jump 3-cone drill Shuttle run	Drafted offensive guards had the slowest 10, 20, and 40-yd dash times, slowest 3-cone drill and shuttle run times and the shortest vertical and broad jump compared to all other player positions
Miller et al (Miller et al., 2010)	Defensive tackle	Weight 10-yd dash	83.1% of drafted defensive tackles had a weight>299 lbs and a 10-yd split time<1.85 seconds
	Defensive end	40-yd dash Weight	94.4% of drafted defensive ends had a 40-yd dash time<4.88 seconds and a weight>254 lbs
	Inside linebacker	20-yd dash Bench press	85% of drafted inside linebackers had a 29-yd split time<2.81 seconds and completed>20 bench press reps
	Outside linebacker	10-yd dash 20-yd shuttle	88.2% of drafted outside linebackers had a 10-yd split time<1.86 seconds and a 20-yd shuttle<4.36 seconds
	Cornerback	40-yd dash 20-yd shuttle	91.9% of drafted cornerbacks had a 40-yd dash time<4.59 seconds and a 20-yd shuttle run time<4.33 seconds
	Safety	40-yd dash 20-yd dash	86% of drafted safeties had a 40-yd dash time<4.59 seconds and a 20-yd dash split time<2.65 seconds

Table 4. Characteristics of the studies evaluating the outcome of the draft and future performance.

Authors	Draft variable	Key findings
Teramoto et al (Teramoto et al., 2016)	10-yd dash	10-yd dash for running backs seems to be the most important in predicting future performance in the NFL as assessed by rushing yards per attempt
	Vertical jump	Vertical jump for wide receivers seems to be the most important in predicting future performance in the NFL as assessed by receiving yards per receiver
	Height	Most important predictor of yards per receiver in wide receivers
Mulholland and Jensen (Mulholland et al., 2014)	Bench press	Predictive of NFL draft order but not important in predicting NFL career success for tight ends
	Broad jump	Predictive of NFL career success but not draft order fo rtight ends
	40-yd dash	Predictive of both NFL draft position and career score for tight ends
Lyons et al (Lyons et al., 2011)	40-yd dash	Significantly related to career performance for 3 consecutive years for allpositions except kickers, punters and offensive linemen
Wolfson et al (Wolfson et al., 2011)	40-yd dash Vertical jump 3-cone drill	Combine statistics were not associated with # of games played, playing in≥1, or playing in ≥ 16 NFL games for QBs
Berri and Simmons (Berri & Simmons, 2011)	Height Wonderlic 40-yd dash	No impact on future NFL performance but dominated draft decision-making and where a quarterback was chosen in the draft/overall draft pick as taller, smarter and faster QBs who played at Division I schools were picked higher in the draft
Kuzmits and Adams (Kuzmits & Adams, 2008)	Vertical Jump	Significant correlations to draft order for QBs but no other measure of future success
	Broad jump	Significant correlations to draft order for QBs but no other measure of future success
	40-yd dash	Predictive of NFL career success for running backs Slower 40-yd dash time correlates with higher 3rd year salary for wide receivers Higher year 1 quarterback ratings are associated with slower times in the 40-yard dash
	10 and 20-yd dash	Correlate with draft order for wide receivers and NFL success for running backs
	Wonderlic	No relationship to NFL success for quarterbacks, running backs or wide receivers
Pitts and Evans (Pitts & Evans, 2018)	Height 40-yd dash Wonderlic	Important implications for draft pick, games played per season, yards per season and approximate value per season for QBs but no statistically significant evidence that a QB's Wonderlic score improves his draft position

in height correlated to a 7.2% lower chance of being on the 5-year roster. The 20-yard shuttle drill was also significantly associated with 5-year roster status for safeties (Pollock et al., 2021). With regards to medical injuries, Brophy et al (Brophy et al., 2011) found that a history of shoulder stabilization significantly reduced the length of career in terms of years and games played. Linemen and linebackers with a history of shoulder stabilization had a significantly shorter career, assessed by years and games played, compared to controls by 30%. While isolated ACL reconstruction did not reduce a player's mean length of career, isolated meniscectomies, as well as combined ACL and meniscal surgery did (Brophy et al., 2009).

3.6.5. Combination of draft and salary

Only two studies assessed the influence of combine measures on draft pick and salary. Treme and Allen (Treme & Allen, 2011) found 40-yard dash times to have statistically (and economically) significant effects on draft order and salary for wide receivers. Meanwhile, Mirabile (Mirabile, 2005) found no statistically significant relationship between intelligence (as measured by the Wonderlic test), and compensation nor a statistically significant relationship between intelligence and draft pick number for quarterbacks.

3.6.6. Playing performance

Based on our search criteria, one paper was found to assess the combine's general influence on playing performance. Cook et al (Cook et al., 2020) determined that defensive backs, linebackers, tight ends, and wide receivers who performed statistically better at the combine than their counterparts played more snaps on average over the first two to four years of their career.

4. Discussion

Dissecting the NFL combine, spanning from medical testing and the physical battery to its impact on measures including draft position, salary, playing performance, career longevity, and future performance provides NFL stakeholders with a broad overview of the measures that may (and may not) hold predictive validity. With this information in mind, stakeholders can draw from these various findings to make evidence-informed decisions in their selection of prospects. Players are likewise key stakeholders in combine participation since their performance at the combine is likely to play some role in their draft rank, salary, and performance metrics such as "minutes played" and "games started" which have significant short and long-term implications for team performance (Robbins, 2012).

Despite the surge of data published within the last decade, with a noticeable spike in 2018 and nearly 50% of studies including representative sample sizes surpassing 500 participants, there remain considerable gaps and imbalances in the research efforts. For instance, only one study examined the combine solely with respect to defensive positions, while 11 studies exclusively examined the combine in relation to offensive positions. The risk of such an imbalance may manifest as "assumption making", where knowledge of offensive players is presumed to apply to defensive positions (either rightfully or

Table 5. Characteristics of the studies evaluating the combine and draft position.

Author	Combine Test	Key Findings
Tucker and Black (Tucker & Black, 2021)	Vertical Jump Bench Press Broad Jump	Higher vertical jump, more repetitions in the bench press, and farther broad jump are correlated with better draft position for inside linebackers, offensive linemen, safeties, wide receivers, tight ends, and wide receivers
	40-yd dash	Statistically significant for all positions except for offensive guards, offensive linemen, offensive tackles, defensive tackles, inside linebackers, outside linebackers and punters
Gill and Brajer (Gill & Brajer, 2012)	40-yd dash	A faster 40-yard dash moves a player up in the draft
	Wonderlic scores	Quantitatively more important for QBs than any other position since a 10- point increase in Wonderlic scores moves a QB up in the draft by 31 positions
Hartman (Hartman, 2011)	40-yd dash	Sprint power was the strongest predictor of draft status but was not significant
Robbins (Robbins, 2010)	40-yd dash	40-yd dash and associated split times and the jumping measures best predict draft order
Sierer et al (Sierer et al., 2008)	40-yd dash	Drafted skill players (wide receivers, cornerbacks, free safeties, strong safeties and running backs), drafted big skill players (fullbacks, linebackers, tight ends and defensive ends) and linemen (centres, offensive guards, offensive tackles and defensive tackles) performed better than nondrafted skill players in the 40-yd dash
	Vertical jump	Skill players (wide receivers, cornerbacks, free safeties, strong safeties and running backs) performed better than nondrafted skill players in the vertical jump
	20-yd shuttle	Skill players (wide receivers, cornerbacks, free safeties, strong safeties and running backs) performed better than nondrafted skill players in the 20-yd shuttle
	3-cone drill	Drafted skill players (wide receivers, cornerbacks, free safeties, strong safeties and running backs), drafted big skill players (fullbacks, linebackers, tight ends and defensive ends) and linemen (centres, offensive guards, offensive tackles and defensive tackles) performed better than nondrafted skill players in the 3-cone drill
	Bench press	Linemen (centres, offensive guards, offensive tackles, and defensive tackles) had significant differences in the 225-lb bench press
McGee and Burkett (McGee & Burkett, 2003)	3-cone drill	Most significant predictor of draft round for running backs, wide receivers, and defensive backs
	Height Weight	Most significant predictor of draft round for running backs, wide receivers, defensive backs, offensive linemen and defensive linemen
	40-yd dash	Most significant predictor of draft round for linebackers
	Vertical jump	Most significant predictor of draft round for running backs, wide receivers, defensive backs
	Bench press Broad jump	Best predictor of draft success for offensive and defensive linemen
	10-yd dash split time	Most significant predictor of draft round for running backs, wide receivers, defensive backs, and line-backers
Pitts and Evans (Pitts & Evans, 2019)	40-yd dash	1 SD improvement in 40-yd dash time is correlated with being picked earlier in the draft for running backs, tight ends, and wide receivers
	Weight Height	Correlate with draft position of running backs and tight ends
	Vertical Jump	1 SD improvement in vertical jump is correlated with being selected earlier in the draft for wide receivers
Kraeutler et al (Kraeutler et al., 2018)	Height	Important for first-round draft picks as first-round pick running backs were shorter
	Weight	Important for first-round draft picks as first-round pick defensive ends were heavier
	Vertical Jump	First round pick QBs had a significantly higher vertical jump
	40-yd dash	First round pick centre backs had a significantly faster 40-yd dash

wrongfully), and it may also lead to an unequal distribution of talent and strength on an NFL team's roster since teams can use the plethora of information available for offensive positions to select the strongest offensive players, while defensive positions lack comparative evidence to evaluate and improve selections in this area.

As for research foci, the descriptive data indicate a substantial focus on medical testing and future performance, combine

tests, and draft position. However, far less research has investigated career longevity, salary, and playing performance, which are all considered measures of future success. It is clear we know which physical tests and injuries influence a prospect's draft status and overall pick number, but beyond this, far less is known about how the combine impacts playing performance and success during one's NFL career (Berri & Simmons, 2011; Pitts & Evans, 2018; Wolfson et al., 2011).

Discrepancies were also found across research focusing on the combine tests, draft position, draft, and future performance as well as medical testing and future performance. Despite the consensus that ACL injury and reconstruction led to significantly worse mean draft round pick and draft pick number, Keller et al (Keller et al., 2015) reported that NFL-calibre athletes who fully recovered from ACL injury showed no signs of diminished functional athletic performance while Provencher et al (Provencher et al., 2018) contradicted this by finding that ACL reconstruction impacted performance by leading to a significantly lower snap percentage. Moreover, for those questioning the impact of ACL injury and repair on career longevity, only one paper by Brophy et al (Brophy et al., 2009) had data available for offensive and defensive linemen. Thus, beyond linemen, insufficient data are available to inform decision-making when drafting other player positions who suffered an ACL injury, once again demonstrating the sparsity of data available to reliably utilize medical combine data as a predictive measure of future performance.

While the 40-yard dash was found to be the most significant predictive variable in terms of draft decision-making for nearly all positions, quarterbacks especially (Berri & Simmons, 2011; Gill & Brajer, 2012), there was less consensus surrounding the remainder of the physical battery. Vincent et al (Vincent et al., 2019) reported the combine to hold a mere modest benefit in predicting future success, measured through NFL playing performance metrics such as sacks, for defensive players. Others have recommended restructuring and eliminating some tests altogether. Purported to be a measure of maximal upper body strength, the repetitions performed for the 225-pound bench press, ranging from 16 to 29, make this test a measure of upper body endurance which is why Russell and Hogan (Russell & Hogan, 2018) recommend replacing the bench press test with a seated medicine ball throw to measure maximal upper body strength. In evaluating the broad and vertical jumps, mixed results were found. Robbins et al (Robbins & Goodale, 2012) concluded the broad and vertical jumps were weak measures of strength and power. Russell and Hogan (Russell & Hogan, 2018) echoed these sentiments and made the argument the broad and vertical jumps were redundant and further advised practitioners to focus on solely the broad jump. This contrasts with Robbins (Robbins, 2012) who recommended retaining the vertical jump over the broad jump due to less common variance with other tests in the physical battery. Overall, these findings question the value of having prospects perform *both* the broad and vertical jump.

Likewise, the 20-yard shuttle run and 3-cone drill were reported to be poor measures of agility since they are static measures that are not representative of the dynamic competition environment (Robbins & Goodale, 2012). Russell and Hogan (Russell & Hogan, 2018) as well as Robbins (Robbins, 2012) added to these findings noting the 20-yard shuttle run and 3-cone drill to be redundant and recommended the 20-yard shuttle run over the 3-cone drill since it mimics player movement on the field and has less shared common variance with other tests.

Studies examining draft and future performance concluded that combine tests influence draft decision-making but not future performance (Berri & Simmons, 2011; Kuzmits & Adams, 2008). For instance, despite the fact that a faster 40-yard dash time leads to an earlier draft round pick, once drafted, the 40-yard dash time had no relationship with first-year performance, as measured by receptions, nor did it lead to more games played or games started for wide receivers (Treme & Allen, 2011). Taking financial compensation to be another measure of future success, Mirabile (Mirabile, 2005) concluded that while Wonderlic scores had no bearing on compensation, a player's height, 40-yard dash time and past level of competition did for quarterbacks. Lyons et al (Lyons et al., 2009) supported this finding noting that past performance, rather than combine performance, significantly predicted future NFL performance in the first four seasons after the combine since the content validity of combine tests was found to be relatively low, contributing to its weak effect size on future performance.

With such controversial findings, the question remains as to why the NFL combine continues to be held annually and held in such high regard to so many. Likely the largest factor accounting for this is the fact that the combine is a major media event that brings in a substantial amount of publicity to the league while serving as a source of revenue from sponsorships, advertisements, and ticket and merchandise sales (Lock & Gratz, 1983) The endowment effect, an emotional bias that causes individuals, or in this case, the NFL, to overvalue an object regardless of its market value may also explain the permanence of the combine, despite its inefficiencies, as the time, money and resources invested into running the event annually causes the league to overvalue its predictive validity and overlook its pitfalls.

5. Limitations and future directions

Despite the implications this review holds for NFL scouts, teams, players and other key stakeholders, there are some limitations that must be considered. Since peer-reviewed articles were part of the inclusion criteria, some key information may have been excluded as a great deal of accessible information concerning the NFL combine is shared on social media platforms, websites, podcasts, and other sources not subject to the verification of the peer-review process. Furthermore, with articles using different measures of success to assess the predictive validity of the combine, the lack of standardization made it difficult to report statistical relationships between variables and across different player positions. Despite these limitations, this review highlights the need for future studies to focus on long-term, future performance measures including salary, career longevity and playing performance, which lack substantive, accessible data through the peer-reviewed system if we are to assess the predictive

validity of the combine. Given the fact that there are far more studies that focused on offensive player positions, more research exclusively evaluating the combine and its impact on defensive players is needed to balance the distribution of data. With the 2021 NFL combine being cancelled due to COVID-19, the only time in its 40-year history (Gordon, 2021), future studies would benefit from investigating how decision making uninformed by combine tests impacts the future performance of 2021 draftees as compared to combine participants.

6. Conclusion

With gaps in positional information and inconsistencies in the combine's effect on draft position, medical testing and future performance, and draft and future performance, we cannot confidently conclude that the NFL combine is a valid nor reliable talent identification tool that is predictive of future performance. However, this is not to say that the combine should be disregarded, as some elements of the combine appear to hold value. For example, the 40-yard dash and its associated split times appeared to be predictive of a prospect's draft status and overall pick number (Hartman, 2011; Lyons et al., 2011; Pitts & Evans, 2018; Robbins, 2010; Sierer et al., 2008; Tucker & Black, 2021). The combine also holds value from a medical perspective since certain injuries, such as ACL reconstruction, have been found to reduce career longevity for offensive and defensive linemen which can inform draft decision-making (Keller et al., 2015; Provencher et al., 2018; Brophy et al., 2009; Brophy et al., 2008; Wang et al., 2018). As a result, experts such as Robbins (Robbins, 2012) recommend keeping the vertical jump test, but removing the broad jump due to redundancy, while Russell and Hogan (Russell & Hogan, 2018) and Robbins (Robbins, 2012) advise retaining the 20-yard shuttle run out of the two agility tests due to its representativeness of player movement on the field. Overall, the strongest conclusion we can make is that more research evaluating long-term future performance measures is needed to comprehensively evaluate the efficacy and predictive validity of the NFL combine.

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